

Today's Agenda

Morphological operations

Texture

Common Morphological Operations

Two basic operations

- Erosion
- Dilation

Other operations

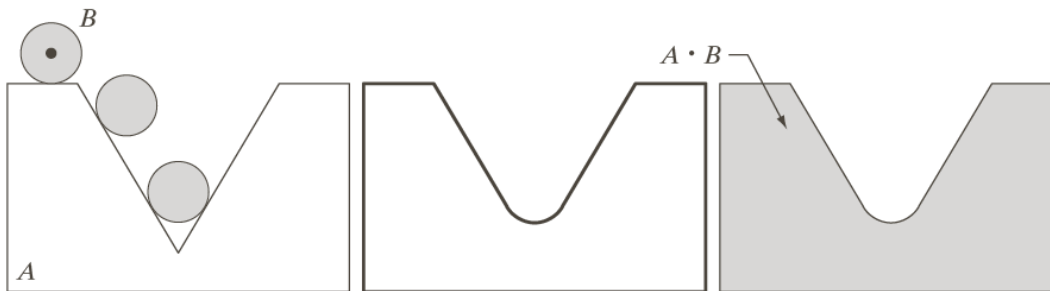
- Opening/closing
- Hit-or-Miss transform
- Thinning/thickening
- Hole filling

Closing

- Smooth the contour of an object
- Fill narrow breaks and gaps
- Eliminate long and thin gulfs
- Eliminate small holes

$$\begin{aligned}
 A \bullet B &= (A \oplus B) \ominus B \\
 (A \bullet B) \bullet B &= A \bullet B \\
 A &\subseteq (A \bullet B) \\
 \text{if } A &\subseteq C, A \bullet B \subseteq C \bullet B
 \end{aligned}$$

Opening and closing are duals of each other $A \bullet B = (A^c \circ \hat{B})^c$

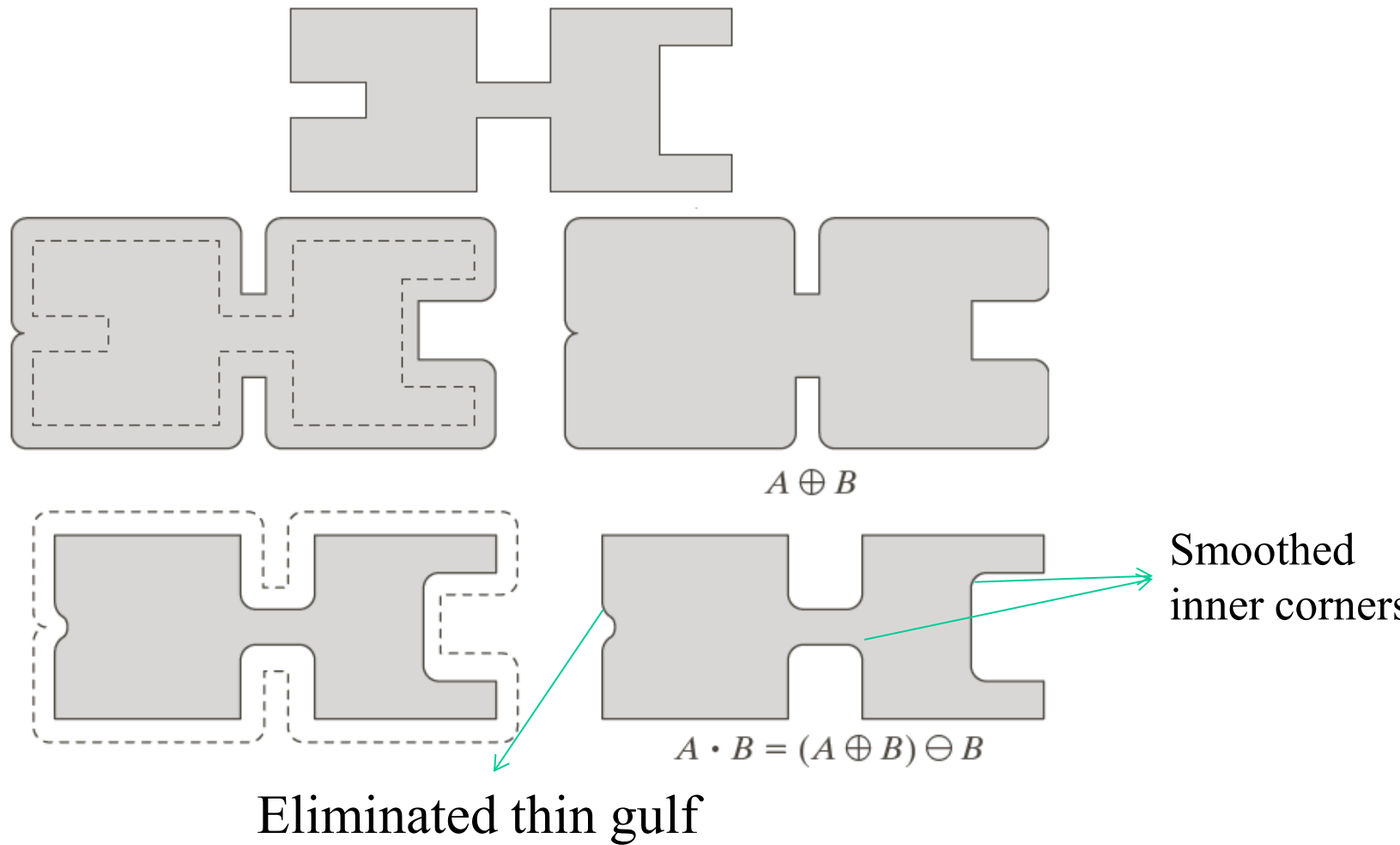


a b c

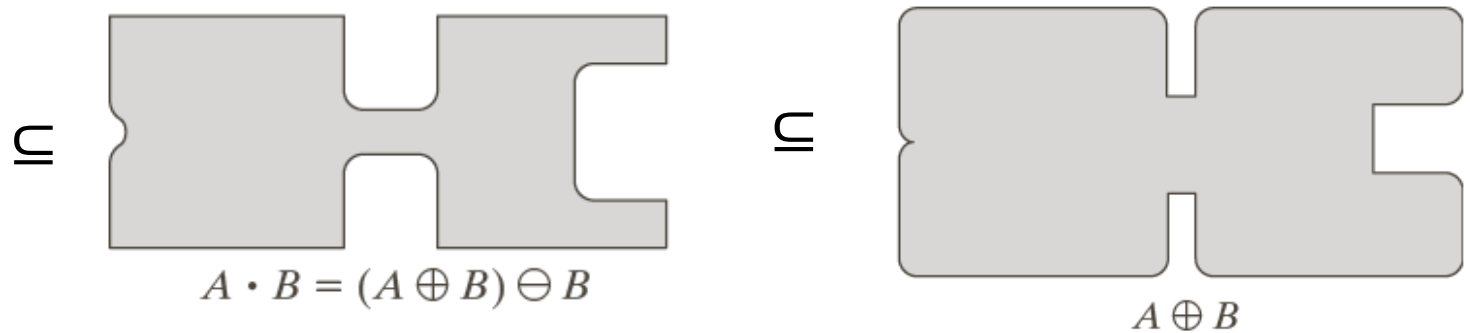
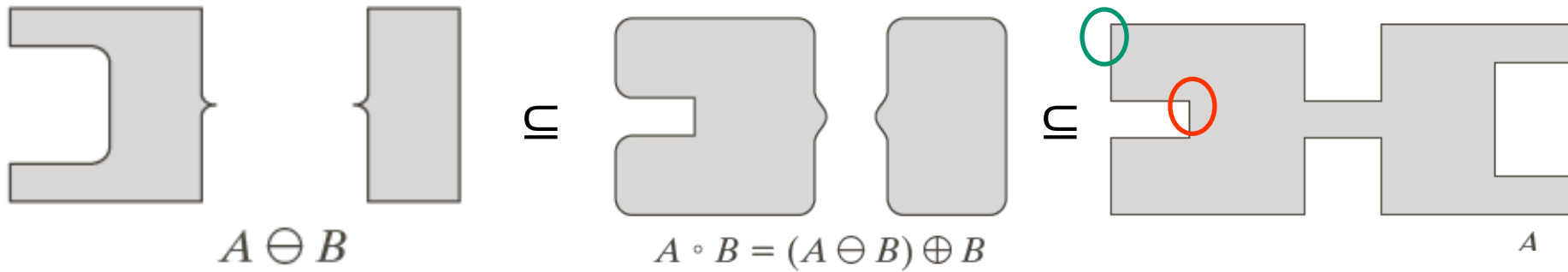
FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

The SE rolls outside the boundary of A .

Closing (Cont'd)



Opening & Closing



$$A \ominus B \subseteq A \circ B \subseteq A \subseteq A \bullet B \subseteq A \oplus B$$

a	b
d	c
e	f

An Example of Opening & Closing

FIGURE 9.11
 (a) Noisy image.
 (b) Structuring element.
 (c) Eroded image.
 (d) Opening of A .
 (e) Dilation of the opening.
 (f) Closing of the opening.
 (Original image courtesy of the National Institute of Standards and Technology.)



- An opening removes all noise
 - removing the white noise by erosion
 - removing the black noise by dilation
- An additional closing fills the gaps

Hit-or-Miss Transform

Shape detection: find D

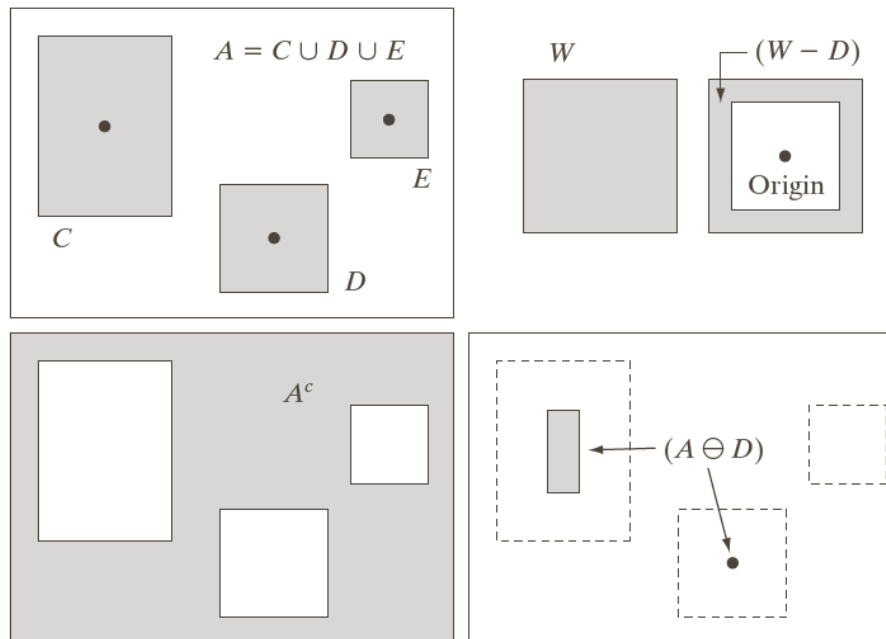
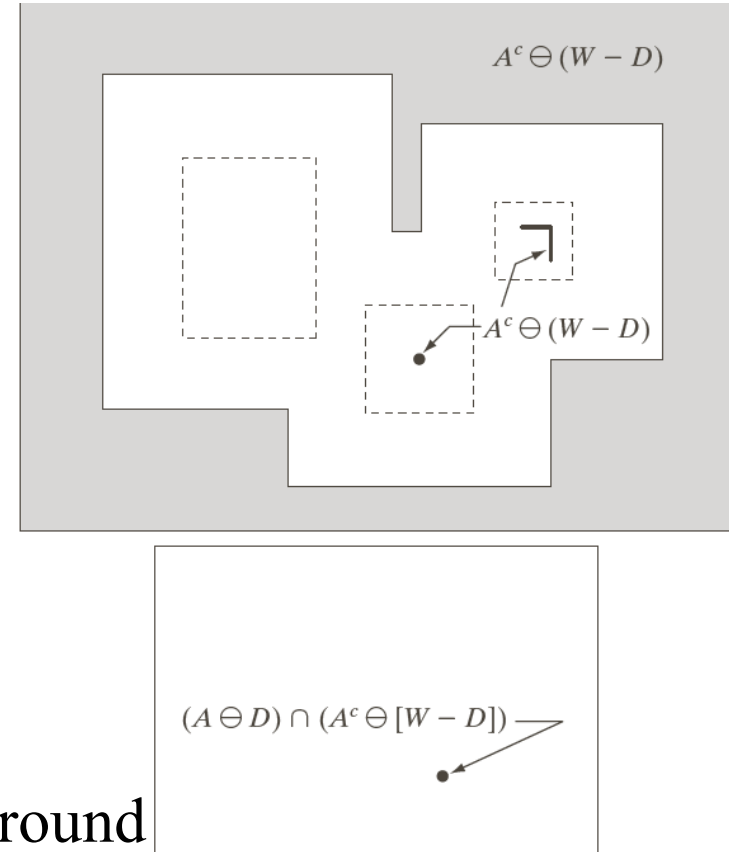


FIGURE 9.12

(a) Set A . (b) A window, W , and the local background of D with respect to W , $(W - D)$. (c) Complement of A . (d) Erosion of A by D . (e) Erosion of A^c by $(W - D)$. (f) Intersection of (d) and (e), showing the location of the origin of D , as desired. The dots indicate the origins of C , D , and E .



Define a set B consisting of D and its background

$$B = (D, W - D) \quad \longrightarrow \quad A * B = (A \ominus D) \cap [A^c \ominus (W - D)]$$

Other Basic Morphological Algorithms

Thinning $A \otimes B = A - (A \circledast B)$

$$B = \{B^1, B^2, B^3, \dots, B^n\} \Rightarrow A \otimes B = \left(\left(\left((A \otimes B^1) \otimes B^2 \right) \dots \right) \otimes B^n \right)$$

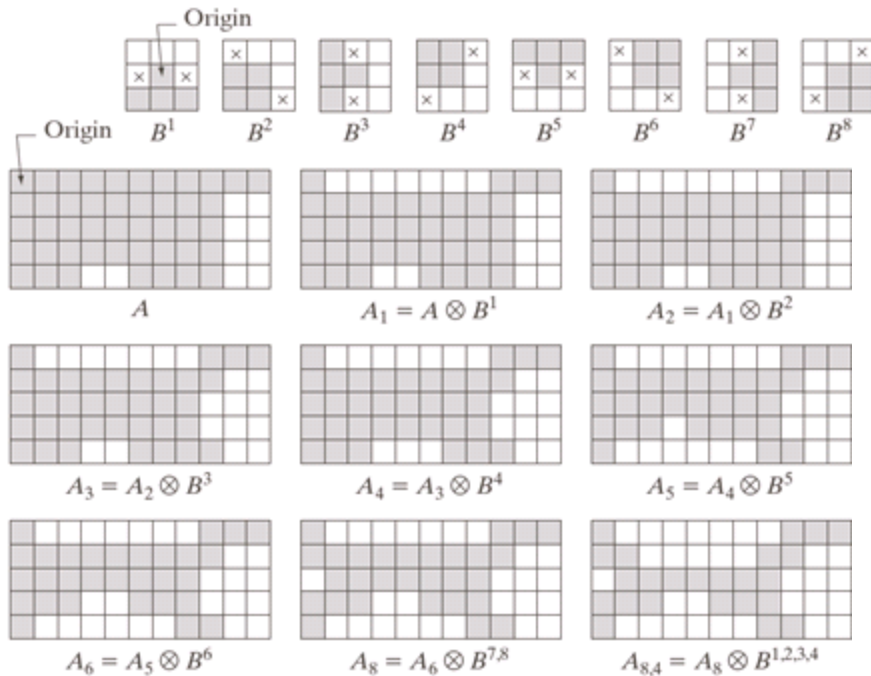


FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A . (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first four elements again. (l) Result after convergence. (m) Conversion to m -connectivity.

a
b c d
e f g
h i j k l m

$A_{8,6} = A_{8,5} \otimes B^6$
No more changes after this.

$A_{8,6}$ converted to
 m -connectivity.

Other Basic Morphological Algorithms

Thickening $A \odot B = A \cup (A * B) \rightarrow$ In practice,
 $B = \{B^1, B^2, B^3, \dots, B^n\}$ $A \odot B = A^c \otimes B$

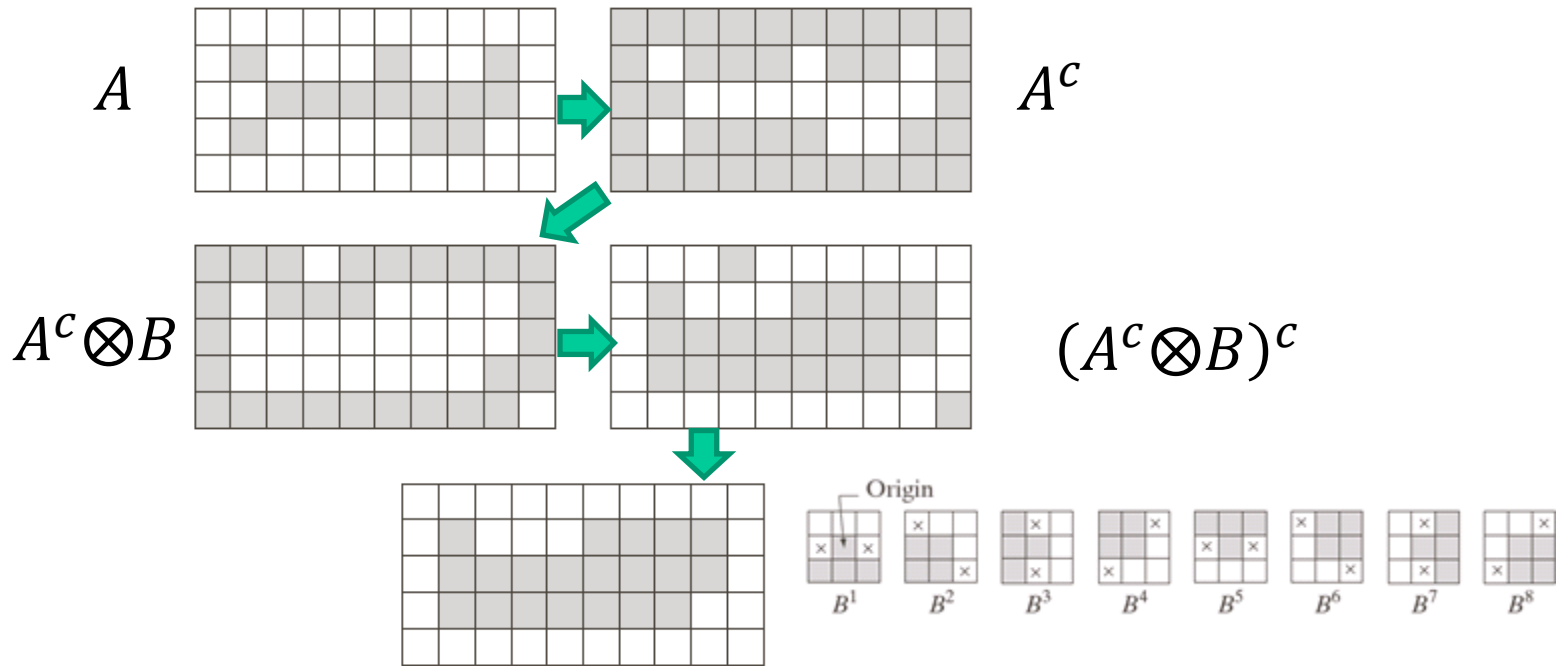


FIGURE 9.22 (a) Set A. (b) Complement of A. (c) Result of thinning the complement of A. (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.

Applications of Morphological Operations

- **Boundary extraction**
- **Hole filing**
- **Connected component analysis**
- **Convex hull extraction**
- **Skeleton analysis**

Basic Morphological Algorithms

Boundary extraction $\beta(A) = A - (A \ominus B)$



a b

FIGURE 9.14

(a) A simple binary image, with 1s represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

Hole Filling

Hole: a background region surrounded by a connected foreground pixels.

Objective: given a point in a hole, fill the hole with foreground pixels.

X_0 is a set of all 0s except the selected background point

A is the set of foreground boundary

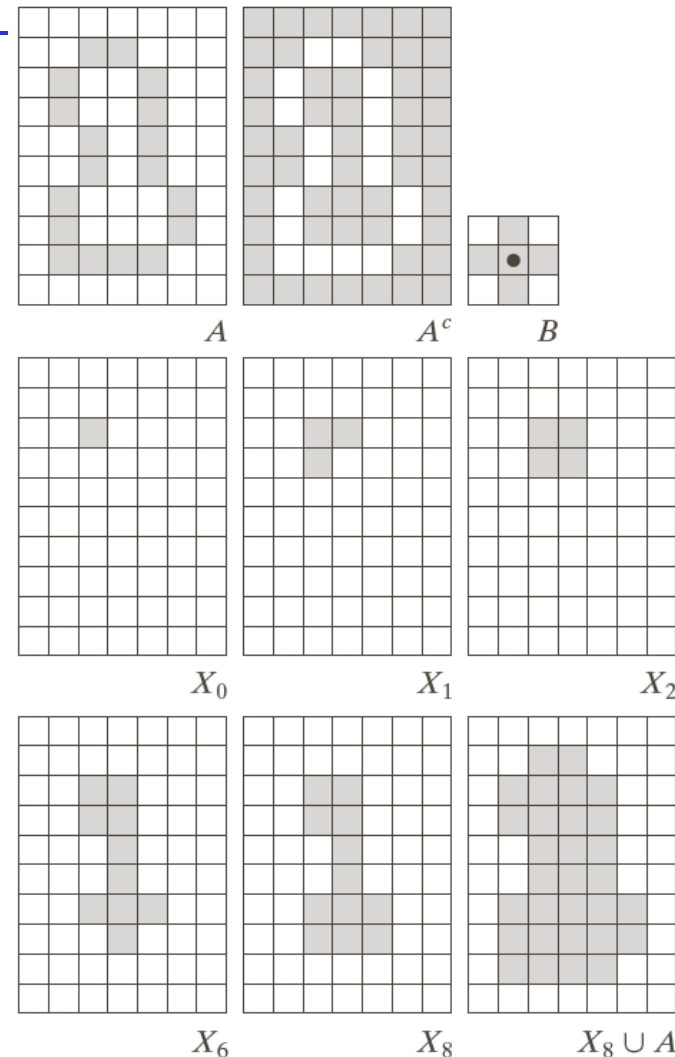
Repeat:

$$X_k = (X_{k-1} \oplus B) \cap A^c \quad k=1,2,3,\dots$$

Until

$$X_k = X_{k-1}$$

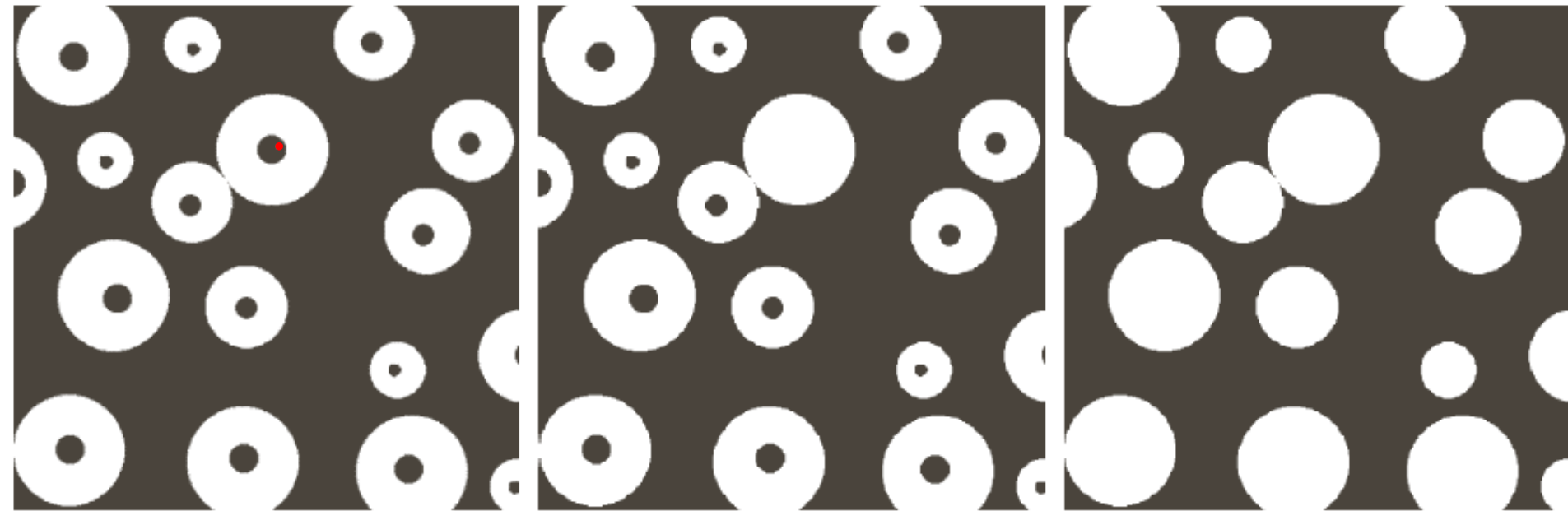
Conditional dilation



a	b	c
d	e	f
g	h	i

FIGURE 9.15 Hole filling. (a) Set A (shown shaded). (b) Complement of A . (c) Structuring element B . (d) Initial point inside the boundary. (e)–(h) Various steps of Eq. (9.5-2). (i) Final result [union of (a) and (h)].

Example



a b c

FIGURE 9.16 (a) Binary image (the red dot inside one of the regions is the starting point for the hole-filling algorithm). (b) Result of filling that region. (c) Result of filling all holes.

Connected Component Analysis

Blob extraction or region labeling

Objective: find connected components in a binary image.

Applications: finding candidates of target object for recognition

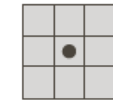
X_0 is a set of all 0s except the selected point belong to the connected component

A is the set containing one or more connected components

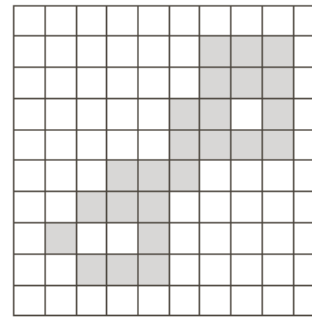
Repeat:

$X_k = (X_{k-1} \oplus B) \cap A \quad k=1,2,3,\dots$

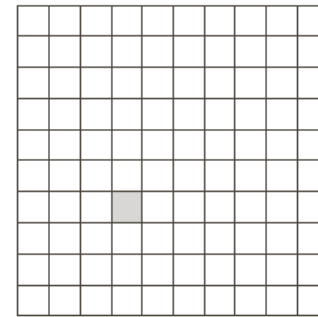
Until $X_k = X_{k-1}$



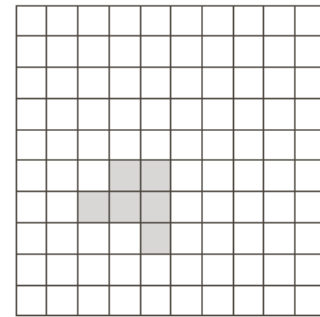
How to select B?



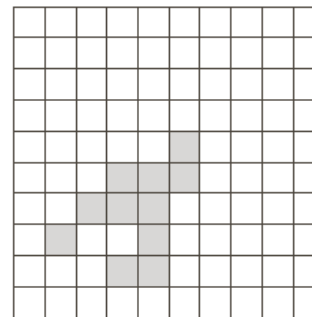
A



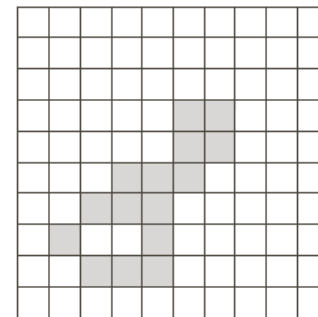
X_0



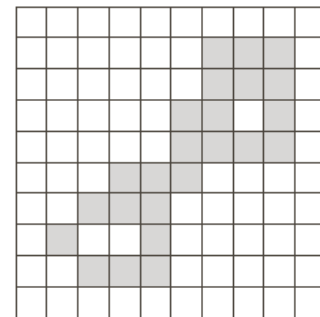
X_1



X_2

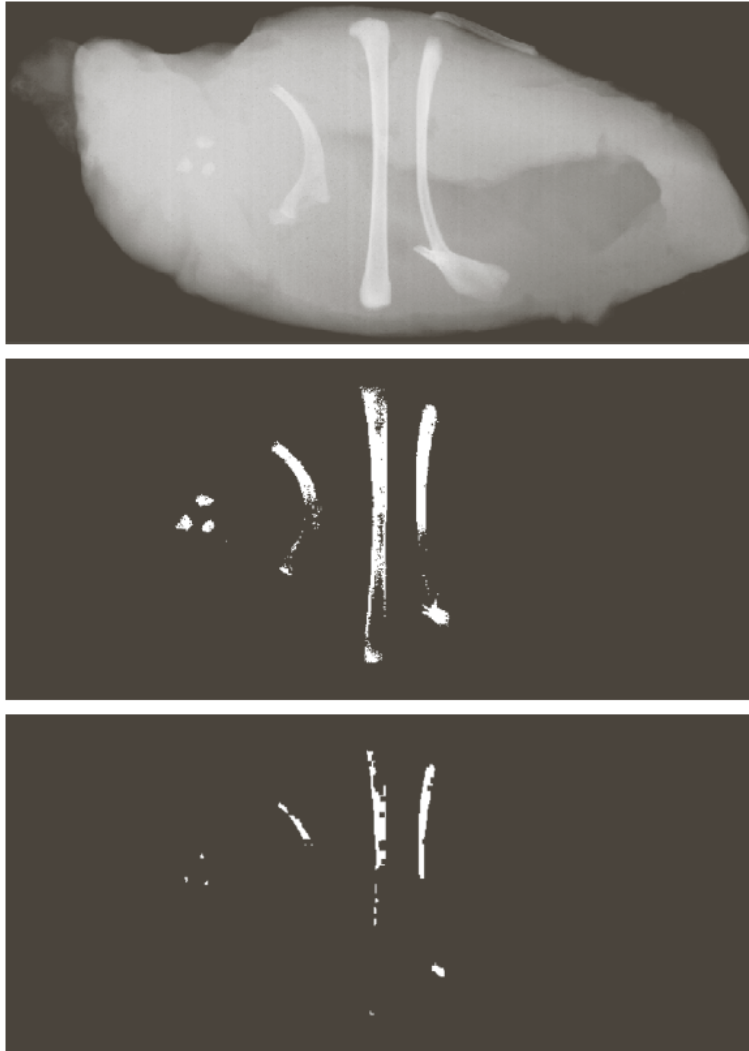


X_3



X_6

Example



a
b
c d

FIGURE 9.18
(a) X-ray image of chicken file with bone fragments.
(b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1s. (d) Number of pixels in the connected components of (c). (Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Convex Hull

Convex: a set **A** is convex if the line segment connecting any two points in **A** is entirely belong to **A**

Examples: rectangle, triangle, circle **are** convex

Ring, hand, many other objects with dents or hollows **are not** convex

Convex hull of a set S: the minimal convex set containing **S**

Applications of finding convex hull: an abstract representation for high level image understanding

Extract Convex Hull

A is the target object and $X_0^i = A$

For each structure element $B^i, i = 1, 2, 3, 4$

Repeat:

$$X_k^i = (X_{k-1}^i \circledast B^i) \cup A$$

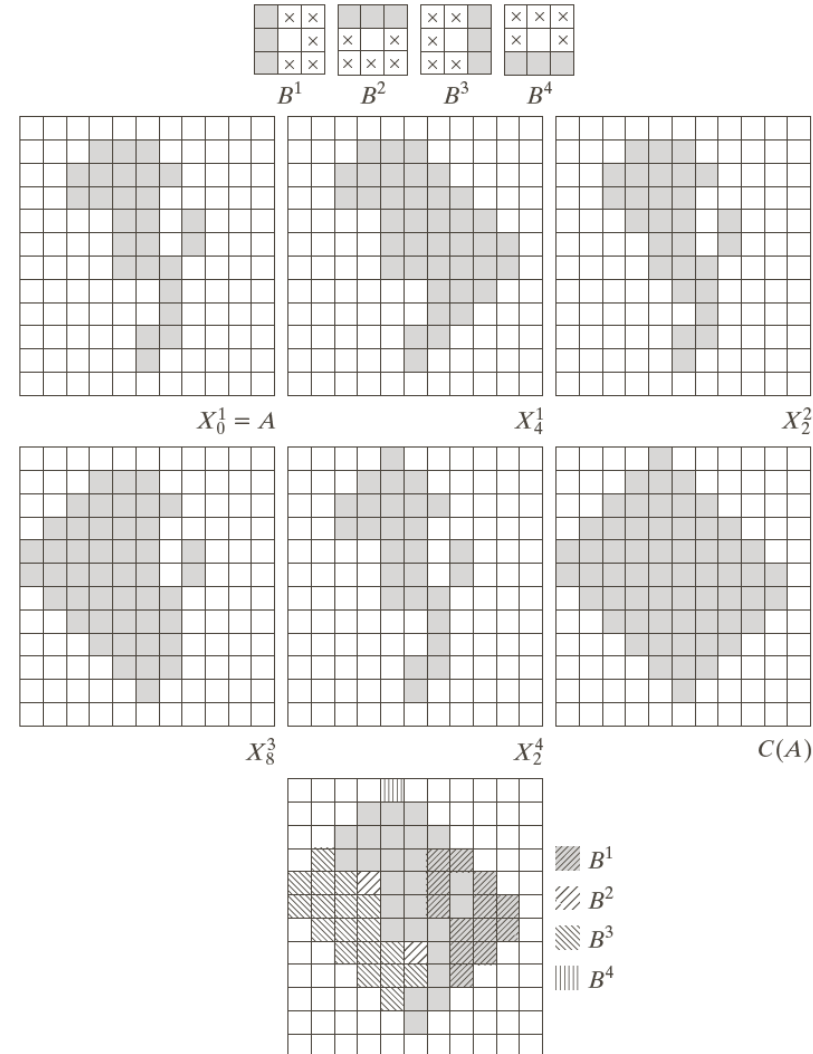
Until

$$X_k^i = X_{k-1}^i$$

Hit-or-miss

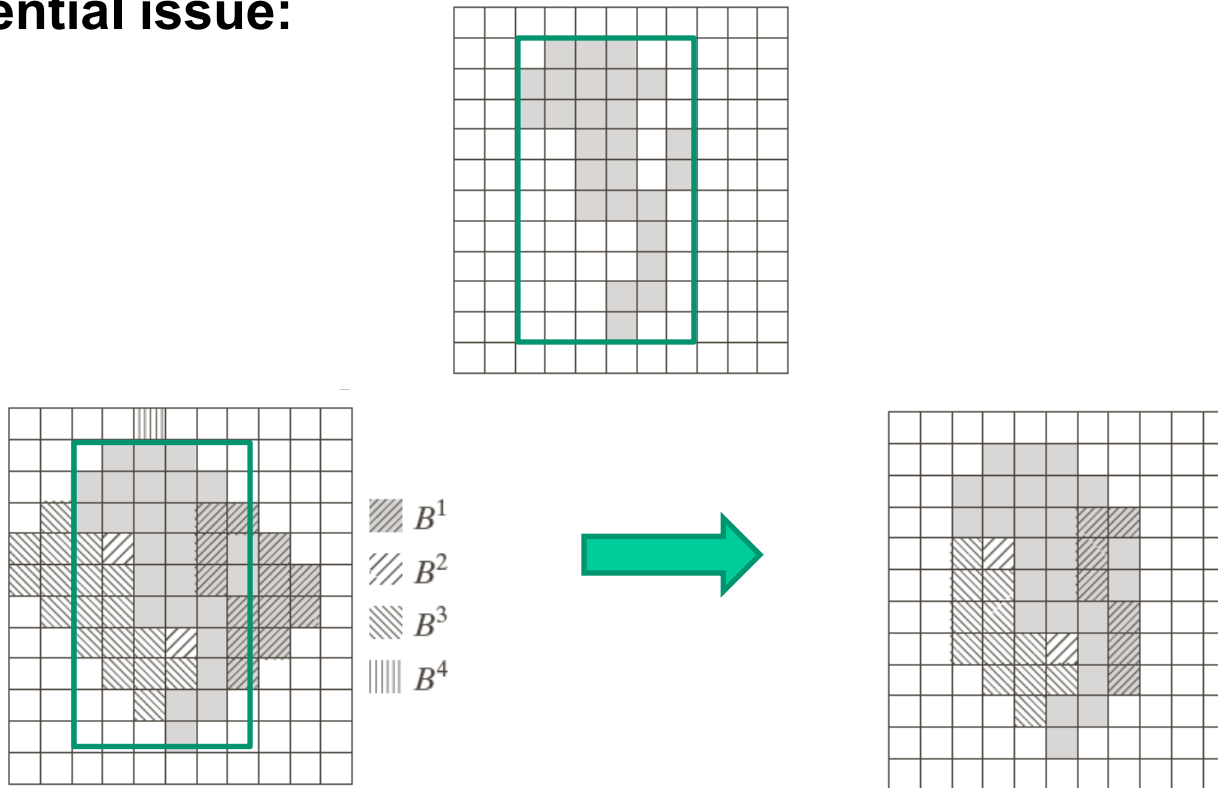
The convex hull of A is

$$C(A) = \bigcup_{i=1}^4 X_k^i$$



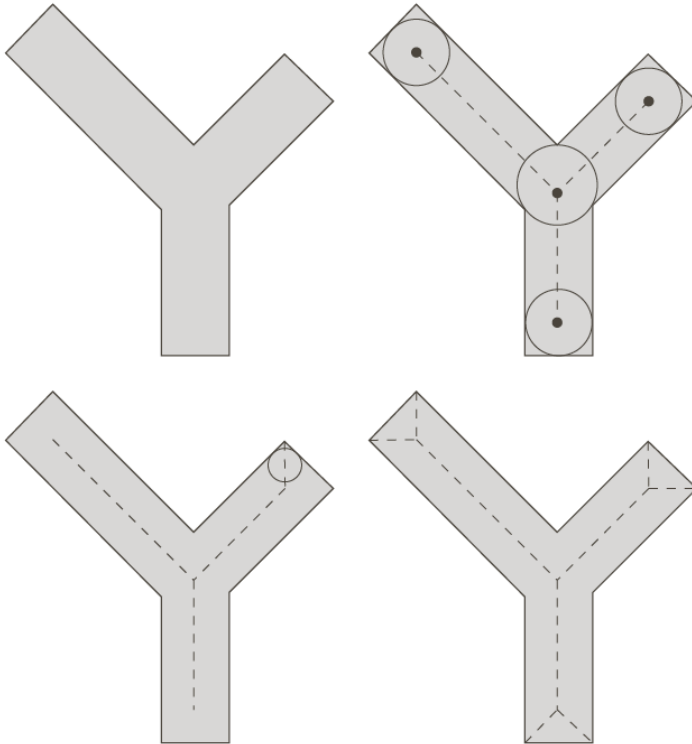
Extract Convex Hull (Cont'd)

Potential issue:



Morphological Skeleton

Skeleton (Defined by centers of maximal disks):



If a point Z belongs to the skeleton of A , we can find a maximal disk that entirely lies in A and touches the boundary of A at no less than two positions.

Applications: an abstract shape representation for high level image understanding, e.g. Optical Character Recognition (OCR)

Morphological Skeleton

$$S(A) = \bigcup_{k=0}^K S_k(A)$$



The k^{th} erosion

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

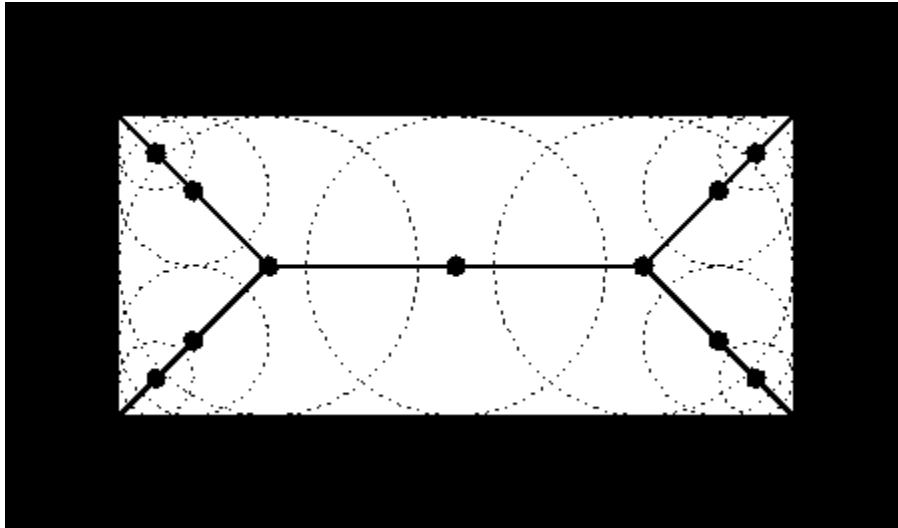
$$K = \max\{k | (A \ominus kB) \neq \emptyset\}$$

Reconstruct A from its skeleton

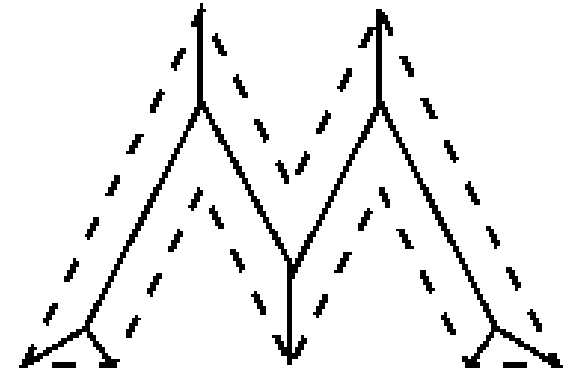
$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

k	$A \ominus kB$	$(A \ominus kB) \circ B$	$S_k(A)$	$\bigcup_{k=0}^K S_k(A)$
0				
1				
2				

Examples



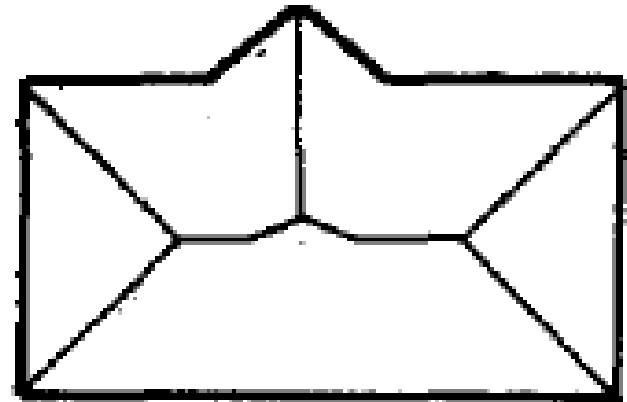
<http://homepages.inf.ed.ac.uk/rbf/HIPR2/skeleton.htm>



Potential issues with skeleton?

Sensitive to noise

Maragos and Schafer, "Morphological Skeleton Representation and Coding of Binary Images", IEEE Trans. on Acoustics, Speech, and Signal Processing, Vol. 34, No. 5, 1986.



Texture

What are the main difference between different textured surfaces?

Change in

- pattern elements
- repetitions



FIGURE 6.1: Although texture is difficult to define, it has some important and valuable properties. In this image, there are many repeated elements (some leaves form repeated “spots”; others, and branches, form “bars” at various scales; and so on). Our perception of the material is quite intimately related to the texture (what would the surface feel like if you ran your fingers over it? what is soggy? what is prickly? what is smooth?). Notice how much information you are getting about the type of plants, their shape, the shape of free space, and so on, from the textures. *Geoff Brightling © Dorling Kindersley, used with permission.*

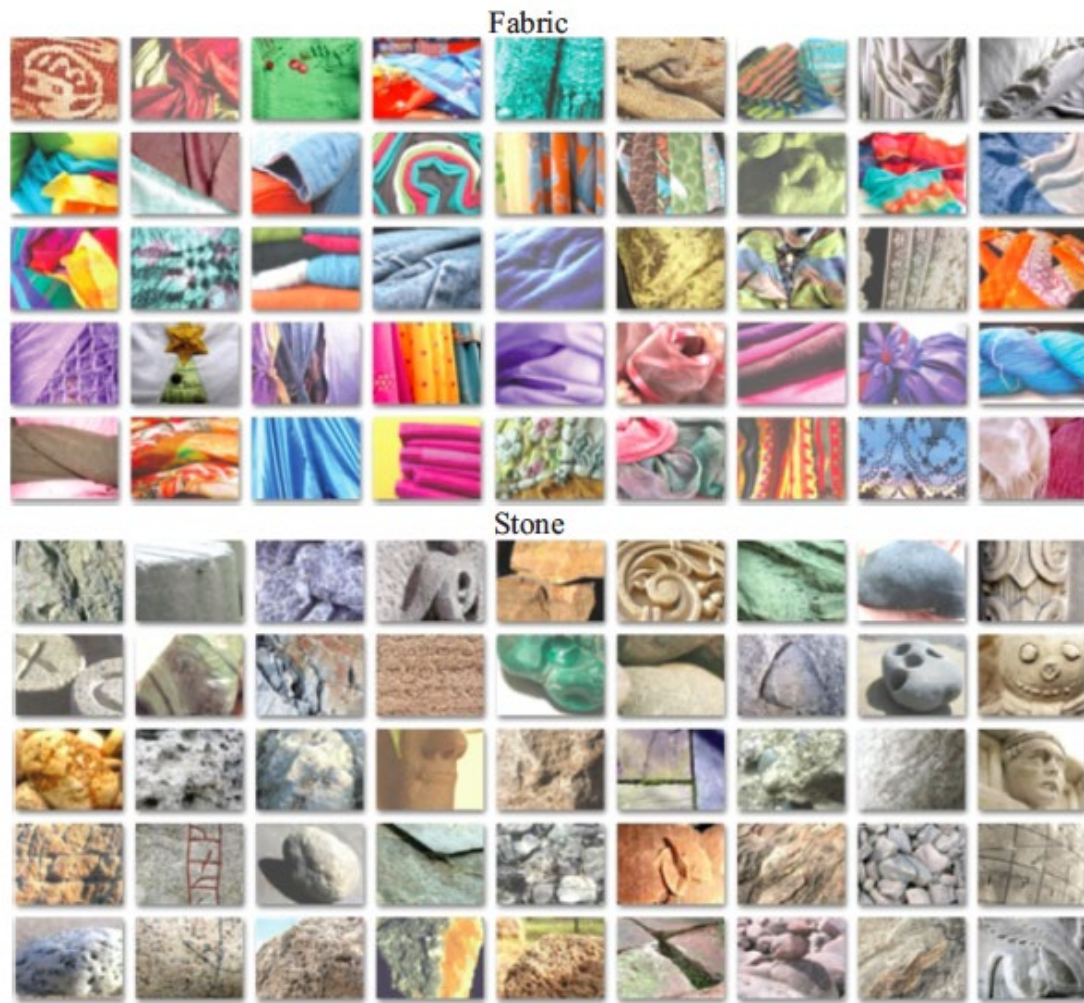


FIGURE 6.2: Typically, different materials display different image textures. These are example images from a collection of 1,000 material images, described in by Sharan *et al.* (2009); there are 100 images in each of the ten categories, including the two categories shown here (fabric and stone). Notice how (a) the textures vary widely, even within a material category; and (b) different materials seem to display quite different textures. This figure shows elements of a database collected by C. Liu, L. Sharan, E. Adelson, and R. Rosenholtz, and published at http://people.csail.mit.edu/lavanya/research_sharan.html. Figure by kind permission of the collectors.

Forsyth and Ponce,
“Computer Vision – A
Modern Approach 2e”

Texture

Patterns of structure from

- changes in surface albedo (e.g., printed cloth)
- changes in surface shape (e.g., bark)
- many small surface patches (e.g., leaves on a bush)

Texture tells us

- what a surface is like
- (sometimes) object identity
- (sometimes) surface shape

Texture

Core problems:

- Texture segmentation
- Texture based recognition
 - Objects, materials, textures
- Texture synthesis
 - Create synthetic images, fill in holes, image editing using computer graphics
- Shape from texture

Key issue: representing texture

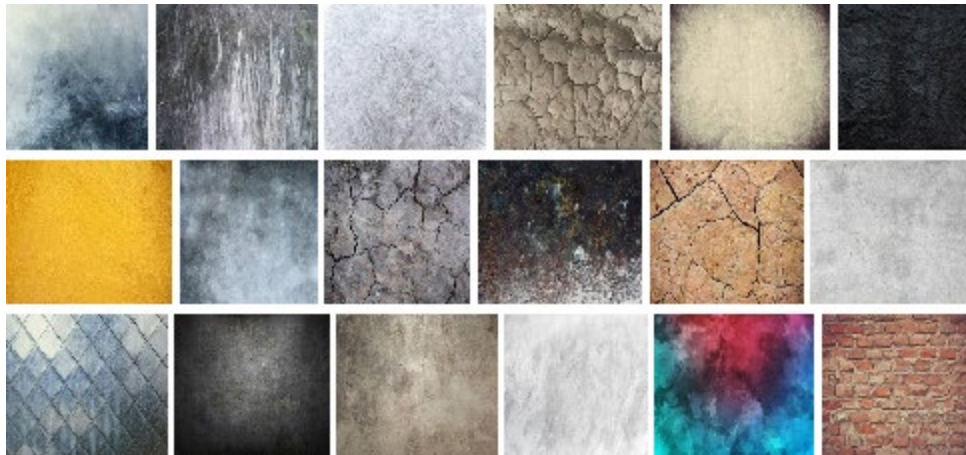
Representing Textures

Textures are made up of

- stylized subelements
- spatially repeated in meaningful ways

Representation:

- find the subelements
- represent their statistics



Examples of texture images by a google search